Rx

Pharmacy Management System

Complete User Guide & Step-by-Step Manual

Flask + SQLite | Bootstrap 5 | Python

Login System | Dashboard | Medicines | Sales | Alerts | Reports

All pages include visual form mockups, navigation instructions, and alert resolution steps

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Section 1 — Getting Started: Login & Navigation

How to access the system and navigate between pages

Step 1 — Start the Application

1. **Open terminal / command prompt in the project folder**

2. **Run: python app.py**

   The server starts at http://127.0.0.1:5000

3. **Open your browser and go to: http://127.0.0.1:5000**

Step 2 — Login Screen

Login — http://127.0.0.1:5000/login

Username *
admin

Password *
••••••••

Login

Role	Username	Password	Access Level
Admin	admin	admin123	Full access — all pages
Pharmacist	pharmacist	pharma123	Medicines + Sales + Reports
Staff	staff	staff123	Sales only

Step 3 — Navigation Bar

The green top navigation bar is always visible after login. Click any item to jump to that page. The Alerts link shows a red badge with the total count of expired/low-stock items.

To log out: click your username at the top right → then click Logout.

Section 2 — Dashboard

Statistics overview, charts, and alert banners

What you see on the Dashboard

Total Medicines	Total Stock Units	Inventory Value	Total Sales
42	1,840	$12,450.00	$3,220.00

Alert Banners — appear automatically when issues exist

✗ 3 Expired! — Medicines past expiry date → Click 'View' to go to Alerts page

! 5 Expiring Soon — Within next 30 days → Click 'View' to review and update

i 2 Low Stock — Below 10 units threshold → Click 'View' to restock

Charts on the Dashboard

Chart	Location	What it shows
Sales Line Chart	Left panel	Daily sales revenue for the last 7 days
Stock Doughnut Chart	Right panel	Normal stock vs Low Stock vs Expiring (coloured)
Recent Medicines Table	Bottom left	Last 5 medicines added with status badges
Activity Log	Bottom right	Last 8 actions (login, add, edit, delete, sale)

Section 3 — Medicines Page

Add, view, edit, delete, search and export medicines

Navigation — http://127.0.0.1:5000/medicines

Click **Medicines** in the top navigation bar. You will see the full inventory table with search, filter, and action buttons.

How to Add a New Medicine

1 **Click the green '+ Add Medicine' button (top right of the page)**

Only Admin and Pharmacist roles see this button

2 **The Add Medicine modal (popup form) appears on screen**

Add Medicine Form

Medicine Name *	Company Name
Paracetamol 500mg	Sun Pharma
Medicine Type	Ref No
Tablet	REF-001
Lot No	Issue Date (YYYY-MM-DD)
LOT-2024-01	2024-01-15
Expiry Date (YYYY-MM-DD) *	Price
2027-12-31	25.00
Quantity (Stock) *	Barcode
100	8901234567890

Save Medicine

! Required fields: Medicine Name, Expiry Date, Quantity. Expiry Date format must be YYYY-MM-DD (e.g. 2027-06-30). Set Quantity above 10 to avoid Low Stock alert.

3 **Click 'Save Medicine' — a green success banner confirms the entry**

The new medicine appears at the top of the table immediately

How to Search Medicines

1. **Type in the Search bar at the top of the Medicines page**

   Search works on: medicine name, company, type, barcode, or ref no

2. **Click Search button OR press Enter**

   Results show instantly. A blue badge shows: Results for: 'paracetamol'

3. **Click 'Clear' to reset and show all medicines again**

How to Edit a Medicine

1. **Find the medicine in the table (use Search if needed)**

2. **Click the pencil (edit) icon in the Actions column**

3. **Update any fields in the Edit form that appears**

   You can fix the Expiry Date or increase Quantity to clear alerts

4. **Click 'Update Medicine' to save changes**

Medicines Table — Row Colour Meaning

Medicine Name	Company	Qty	Expiry Date	Status
Amoxicillin 250mg	Cipla	3	2024-11-01	EXPIRED
Ibuprofen 400mg	Pfizer	8	2025-04-15	Expiring Soon
Paracetamol 500mg	Sun Pharma	150	2027-06-30	OK

Colour	Meaning	Action Required
Red row	Medicine has expired (past today's date)	Edit → update expiry date OR delete
Yellow row	Expiring within 30 days OR qty ≤ 10	Edit → update expiry / restock quantity
White row	Healthy — no issues	No action needed

How to Export to Excel

1. **Click the green 'Export Excel' button (top right of Medicines page)**

   Admin and Pharmacist only

2 **Browser downloads: medicines_2025-03-16.xlsx automatically**

✓ The Excel file uses colour coding: red rows = expired medicines, yellow rows = low stock. All 16 columns are included: name, company, type, lot, expiry, qty, price, barcode, etc.

Section 4 — Sales Page

Record sales transactions and view history

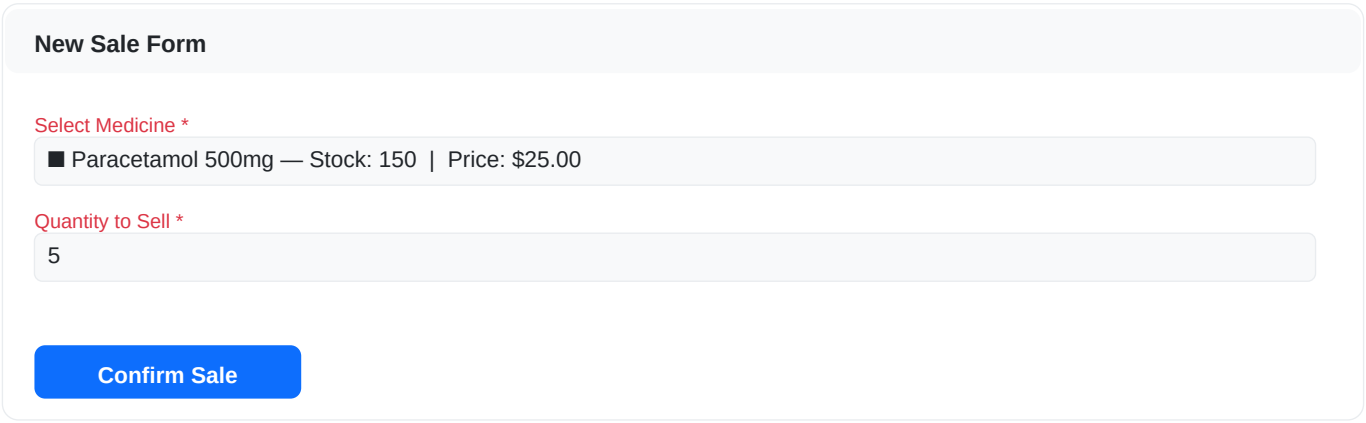
Navigation — http://127.0.0.1:5000/sales

Click **Sales** in the navigation bar. You see all past transactions. All 3 roles (Admin, Pharmacist, Staff) can record sales.

How to Record a New Sale

1 **Click the blue 'New Sale' button (top right)**

2 **The Record Sale modal appears:**



3 **The Total is auto-calculated: Qty × Unit Price = Total**

Example: 5 × $25.00 = $125.00

4 **Click 'Confirm Sale' to save the transaction**

Stock is instantly deducted from inventory

Sales Table — What each column means

#	Medicine	Qty	Unit Price	Total	Sold By	Date
1	Paracetamol 500mg	5	$25.00	$125.00	pharmacist	2025-03-16 10:30
2	Ibuprofen 400mg	2	$18.00	$36.00	admin	2025-03-16 11:00

✗ If the quantity entered exceeds available stock, a warning appears: 'Insufficient stock. Available: X'. The sale will not be saved.

i Sales revenue feeds the Dashboard's '7-day Sales Chart' and 'Total Sales' stat card automatically.

Section 5 — Alerts Page

Identify and fix expired medicines and low stock warnings

Navigation — http://127.0.0.1:5000/alerts

Click **Alerts** in the navigation bar. A red badge shows the total number of issues. This page has 3 sections: Expired, Expiring Soon, Low Stock.

Expired Medicines	Expiring Within 30 Days	Low Stock Items (≤ 10 units)
3	5	4

Section A — Expired Medicines (Red)

✗ These medicines have an expiry date in the past. They show in a red table with 'Immediate action required' badge at the top.

1. **Click 'View' button next to any expired medicine**

   This opens the Medicine Detail page

2. **On the detail page, click the 'Edit' button**

3. **Change the Expiry Date to a future date (e.g. 2027-12-31)**

   Format must be YYYY-MM-DD

4. **Click 'Update Medicine' — red alert disappears immediately**

★ Alternatively: If the medicine is genuinely expired and should be removed, go to Medicines page, find it, and click the trash (delete) icon. Admin only.

Section B — Expiring Soon (Yellow)

! Medicines whose expiry date falls within the next 30 days. A 'Days Left' column shows exactly how many days remain.

1. **Click 'View' to open the medicine detail**

2. **Note the days remaining — plan to use or replace the stock**

3. **If restocked with a new batch: edit the Expiry Date to the new batch date**

Section C — Low Stock (Blue)

i Any medicine with quantity at or below 10 units. The threshold is set to 10 in app.py (LOW_STOCK_THRESHOLD = 10).

1. **Go to Medicines page, find the medicine**

2. **Click the edit (pencil) icon**

3. **Change Quantity to the new restocked amount (e.g. 100)**

   Enter the total current quantity after restocking

4. **Click 'Update Medicine' — low stock badge disappears**

Section 6 — Reports Page

Analytics, charts, and inventory summaries — Admin/Pharmacist only

Navigation — http://127.0.0.1:5000/reports

! Only Admin and Pharmacist roles can access the Reports page. Staff accounts will see an 'Access denied' message.

Report Section	What It Shows	How to Use
Medicines by Type	Count, total qty & value grouped by type (Tablet, Syrup, etc.)	Understand inventory composition; bar chart visual inc
Top 10 Low Stock	10 medicines with the lowest quantity, sorted ascending	Use as a reorder checklist — edit each to restock
Expiring Soon List	All medicines expiring within 30 days, sorted by date	Plan stock rotation or procurement
30-Day Sales Summary	Daily transactions, units sold, and revenue for last 30 days	Track sales trends day by day
Total Inventory Value	Sum of (price × quantity) for all medicines	Financial overview of your entire stock

Section 7 — Users Page

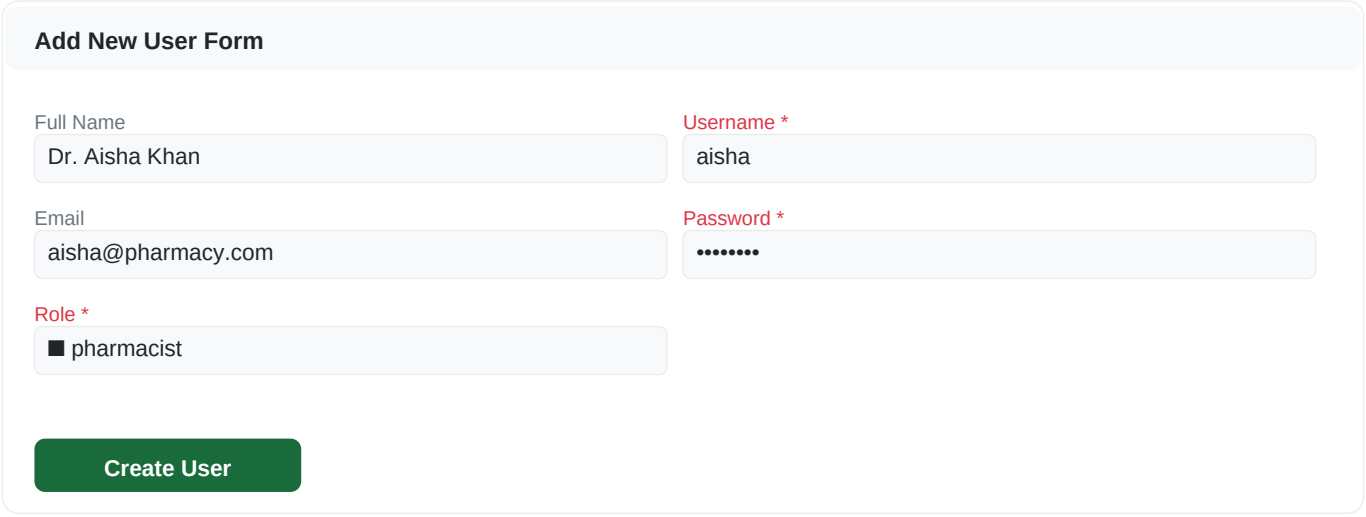
Create and manage staff accounts — Admin only

Navigation — http://127.0.0.1:5000/users

✗ Only the Admin account can access the Users page.

How to Add a New Staff Account

1 **Click '+ Add User' button on the Users page**



Role	Can Add/Edit Medicine	Can Delete Medicine	Can Record Sales	Can View Reports	Can Manage Users
admin	Yes	Yes	Yes	Yes	Yes
pharmacist	Yes	No	Yes	Yes	No
staff	No	No	Yes	No	No

2 **You cannot delete your own account**

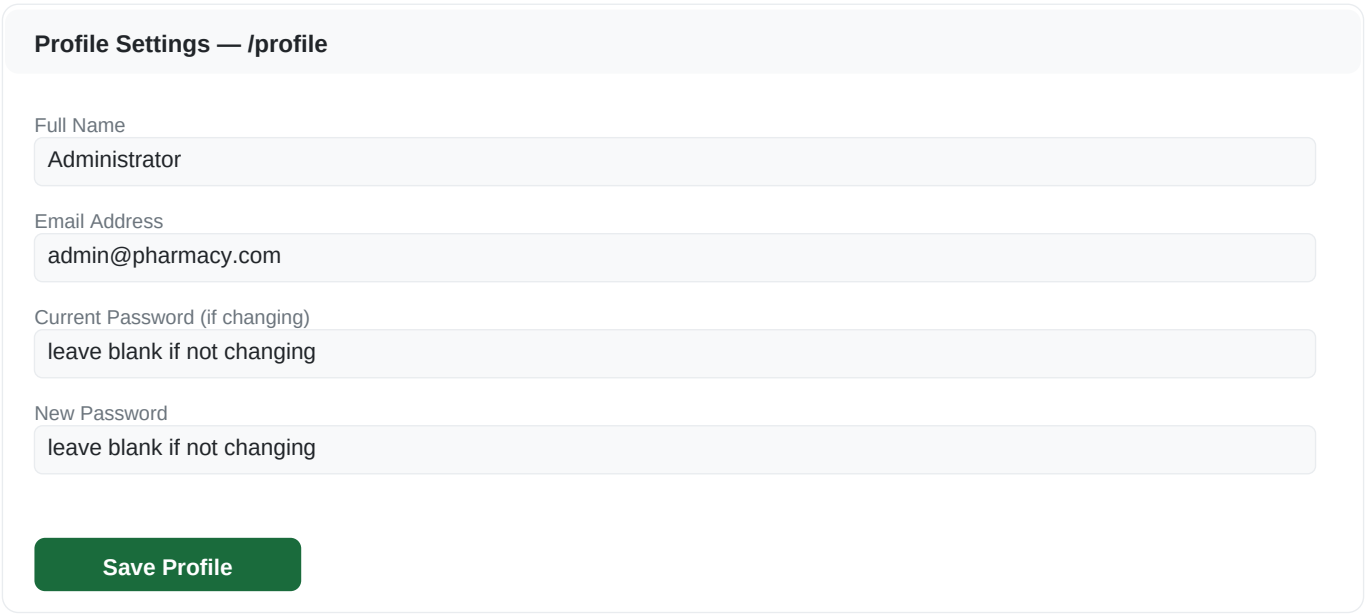
This is a safety feature — another admin must delete your account if needed

Section 8 — Profile Page

Update your name, email, and change your password

Navigation — Click your username (top right) → Profile

Every logged-in user can access their own profile page.



To change your password: fill in both 'Current Password' and 'New Password'. If you leave them blank, only name and email are updated.

Section 9 — Quick Reference: Alert Colour Codes

At-a-glance guide to all visual indicators in the system

Visual Indicator	Location	Meaning	How to Clear It
Red row in table	Medicines page	Medicine is EXPIRED (past today)	Edit → set future Expiry Date
Yellow row in table	Medicines page	Expiring in ≤30 days	Edit → update Expiry Date
Yellow row in table	Medicines page	Quantity ≤ 10 (Low Stock)	Edit → increase Quantity
Red badge on Qty cell	Medicines / Alerts	Critically low stock	Edit → restock quantity
Green badge on Qty cell	Medicines page	Healthy stock level (> 10)	No action needed
Red 'Expired' badge	Status column	Past expiry date	Edit or delete the medicine
Yellow 'Expiring' badge	Status column	Expiring within 30 days	Update expiry date
Green 'OK' badge	Status column	No issues	No action needed
Red number on Alerts nav	Top navigation bar	Total issues count (expired + low stock)	Resolve individual alerts
Red banner on Dashboard	Dashboard top	Medicines have expired	Click 'View' → go to Alerts
Yellow banner on Dashboard	Dashboard top	Medicines expiring soon	Click 'View' → go to Alerts
Blue banner on Dashboard	Dashboard top	Low stock items exist	Click 'View' → go to Alerts

Section 10 — Data Persistence & Seed Data

Will your data be lost when you close the project?

Short Answer: NO — your data is SAFE

✓ All data (medicines, sales, users, logs) is stored in a SQLite database file on disk. Stopping or restarting the Flask server does NOT delete any data.

Where is the database file stored?

File	Location	Contains
pharmacy.db	pharmacy_app/instance/pharmacy.db	All tables: medicines, users, sales, activity_logs

What happens with Seed Data?

The file **seed_data.py** was used once to pre-populate the database with sample medicines, users, and sales. Once that data is in the database file, it persists permanently regardless of whether you close or restart the server.

Event	Data Safe?	Explanation
Close Flask server (Ctrl+C)	YES	Server stops but database file unchanged on disk
Restart Flask server	YES	Server reads the same .db file on startup
Restart your computer	YES	File stays on disk, unaffected
Delete instance/pharmacy.db	NO	This destroys all data — do not delete this file
Run seed_data.py again	DUPLICATES	Re-running seed may add duplicate entries — avoid
Git push / pull	RISK	Add pharmacy.db to .gitignore to avoid overwriting data

Backup Recommendation

★ To back up your data: simply copy the file pharmacy_app/instance/pharmacy.db to a safe location (USB drive, cloud storage, etc.). To restore: replace the file with your backup copy and restart the server.

i The .gitignore file (generated with this guide) excludes pharmacy.db from Git so your production data is never accidentally overwritten by a git pull.